# Assignment: Network Security (15 Questions)

- Differentiate between symmetric and asymmetric cryptography. Explain AES and RSA with use cases.
- What are hash functions? Compare MD5 and SHA, and explain their role in security.
- Explain Public Key Infrastructure (PKI), Digital Signatures, and the role of Certificate Authorities (CAs).
- Describe Kerberos authentication and X.509 certificates in secure communication.
- Explain the CIA triad with practical examples and its importance in network security.
- Describe TCP/IP communication flags and explain port scanning using Nmap.
- Discuss different types of attacks across network layers and their countermeasures.
- Explain IPsec architecture and its components (AH, ESP, IKE).
- Differentiate between Transport mode and Tunnel mode in IPsec.
- Explain TLS architecture, TLS handshake, and common TLS vulnerabilities.
- Explain phishing and BEC attacks. Describe SPF, DKIM, and DMARC mechanisms.
- Discuss secure email protocols (SMTP, IMAP, POP3, STARTTLS) and best practices like MFA.
- Explain ARP spoofing, IP spoofing, and SYN flooding attacks with countermeasures.
- Describe application-level attacks such as SQL Injection, MITM, and DoS/DDoS attacks.
- Firewall types and IDS/IPS
- VPN types and tunneling protocols (OpenVPN, WireGuard)
- Wi-Fi security protocols (WPA2, WPA3)
- How does Man-in-the-Middle Attack differ when performed using ARP spoofing vs DNS spoofing
- Differentiate between IEEE 802.11a, IEEE 802.11b, and IEEE 802.11g based on performance, compatibility, and real-world usage.
- Differentiate between single firewall DMZ and dual firewall DMZ architectures with respect to security, cost, complexity, and risk exposure.
- IoT security challenges and protection mechanisms

**An organization has received several complaints from employees stating that whenever they try to access the company's banking portal, they are being redirected to a fraudulent website that closely resembles the original one. Upon investigation, it was found that the local DNS resolver has been altered, with DNS records pointing the legitimate domain name to an unfamiliar IP address.**

**As a cybersecurity analyst, answer the following:**

a) Identify the type of DNS attack involved in this scenario.

b) Explain step-by-step how the attacker might have carried out this attack.

c) Describe the process through which users were redirected to the malicious website.

d) Recommend at least three methods for detecting and preventing such attacks in the future.

**A university computer lab consists of several systems connected through a network switch. A student observes that while accessing websites, login credentials entered on HTTP sites are being compromised. Further network analysis reveals that one system in the lab is continuously broadcasting forged ARP reply messages to other devices.**

**Assume you are a cybersecurity analyst and answer the following:**

a) Explain how the attacker carried out an ARP spoofing attack in this scenario.

b) Describe how this attack allows the attacker to establish a Man-in-the-Middle (MITM) position between users and the network gateway.

c) Explain the impact of this MITM attack on user data and communication.

d) Suggest at least three preventive measures the university can implement to protect against such attacks in the future.